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**UNIVERSITY OF CAPE TOWN
DEPARTMENT OF ELECTRICAL ENGINEERING**

EEE3094S

Control Systems Engineering

Q1	
Q2	
Q3	
Q4	
Q5	
Q6	
Q7	
Total	

January Examination 2023

TIME: 3 Hours

TOTAL MARKS: 100

INSTRUCTIONS

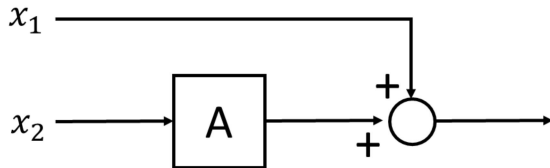
1. Attempt ALL questions.
2. Closed book format.
4. You are allowed one **hand written**, double sided, A4 formula sheet.
3. Hand in all drawings on which you have worked and your formula sheet.
4. Transform tables and Engineering charts are provided at the end of this paper.

Internal Examiners: Prof E Boje, Dr S Shield

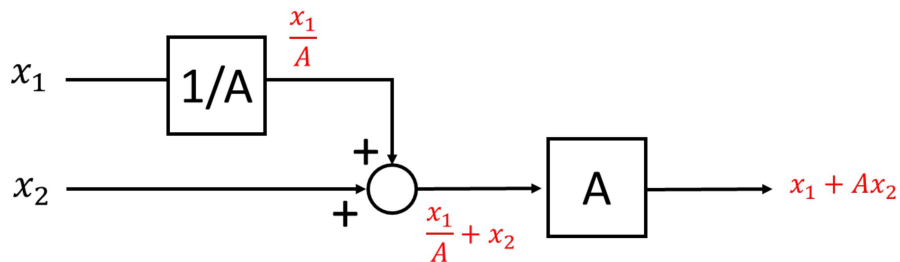
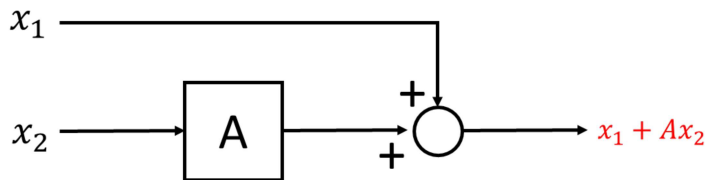
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Question 1 [14 marks]

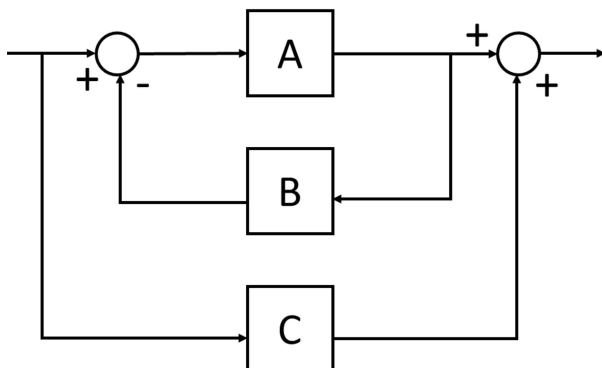
- a) Draw an equivalent of the following block diagram with the summing junction moved to before block A. Indicate the values of the signals at all nodes in terms of x_1 and x_2 to confirm that the systems give the same output. (3)



Solution:



- b) Simplify the following block diagram into a single block. (3)



Solution: A and B are a feedback loop and can be combined into $\frac{A}{1+AB}$

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This is in parallel with C, giving $\frac{A+C}{1+AB}$

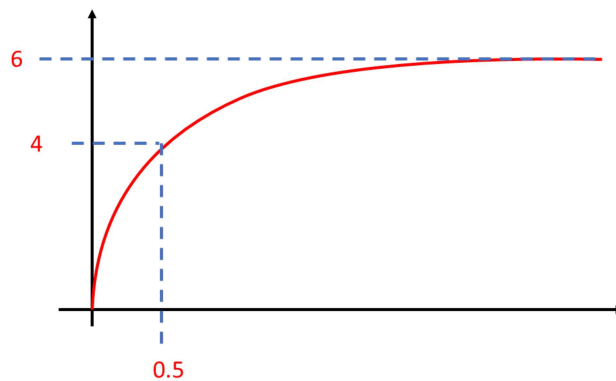
- c) State the type number of the system $L(s) = \frac{3}{s(s+2)}$ and give its error constant with respect to a unit ramp input (in a negative gain feedback configuration). (2)

Solution: This is a type 1 system. The error constant with respect to a unit ramp is $\lim_{s \rightarrow 0} s \frac{3}{s(s+2)} = \frac{3}{2}$

- d) Sketch the unit step response of the system $P(s) = \frac{6}{2s+1}$.

Solution: The gain of the system is 6.

The time constant of the system is 2, so the response will reach 63% of the way to the final value at 0.5 seconds.



(3)

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Question 2 [15 marks]

Consider the Bode plot of a system shown in Figure 2.

- a) Use the plot to determine the following characteristics of the system:
- The gain, A (1)
 - The natural frequency, ω_n (2)
 - The damping ratio, ζ (2)

Solution:

- The gain is 0 dB = 1
- The peak occurs at 10 rad/s
- The magnitude at the peak is 10 dB. Solve $-20 \log(2\zeta) = 10$ to get $\zeta \approx 0.2$

- b) Determine the damped frequency ω_d , rise time t_r , time to peak t_p , peak magnitude M_p and settling time to within 5% of the final value $t_{5\%}$. (5)

Solution:

$$\text{Damped frequency: } \omega_d = 10\sqrt{1 - 0.2^2} = 9.79$$

$$\text{Rise time: } \theta = \arccos(0.2) = 1.37; t_r = \frac{\pi - 1.37}{9.79} = 0.18$$

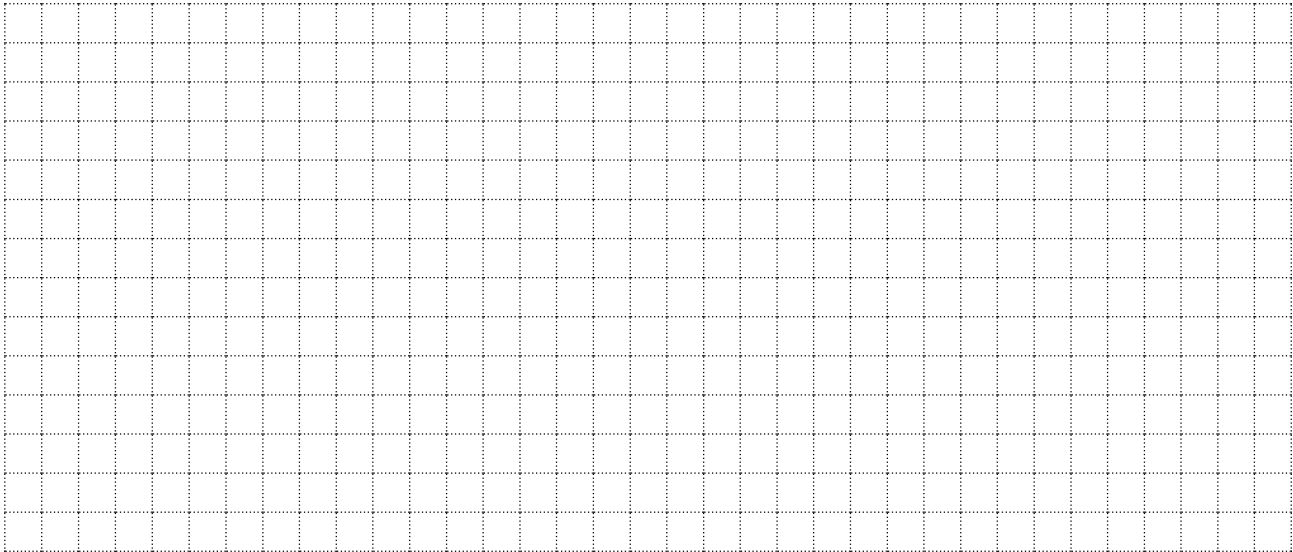
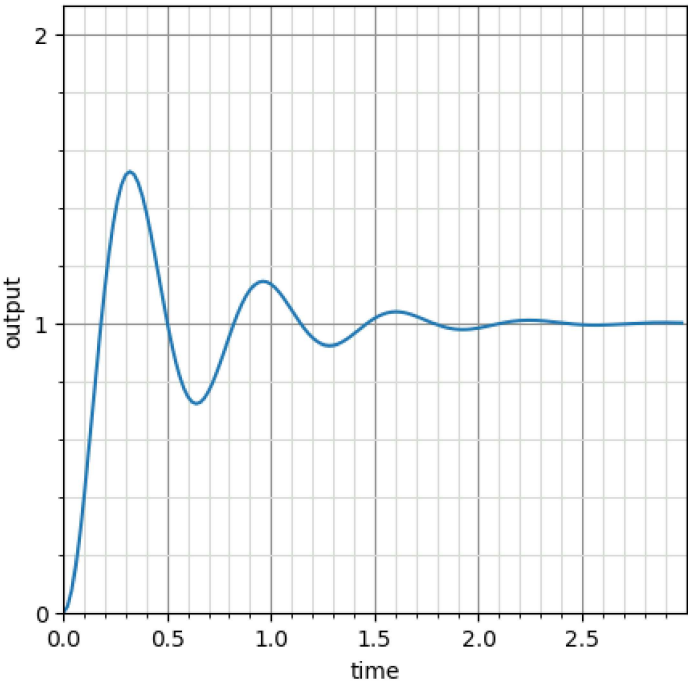
$$\text{Time to peak: } t_p = \frac{\pi}{9.79} = 0.32$$

$$\text{Peak magnitude: } M = 1 + e^{\frac{-0.2\pi}{\sqrt{1-0.2^2}}} = 1.52$$

$$\text{Settling time: } t_{5\%} = \frac{\ln(0.05\sqrt{1-0.2^2})}{-(0.2)(10)} = 1.50$$

- c) Hence, sketch the unit step response of the system, showing the characteristics in (b). (5)

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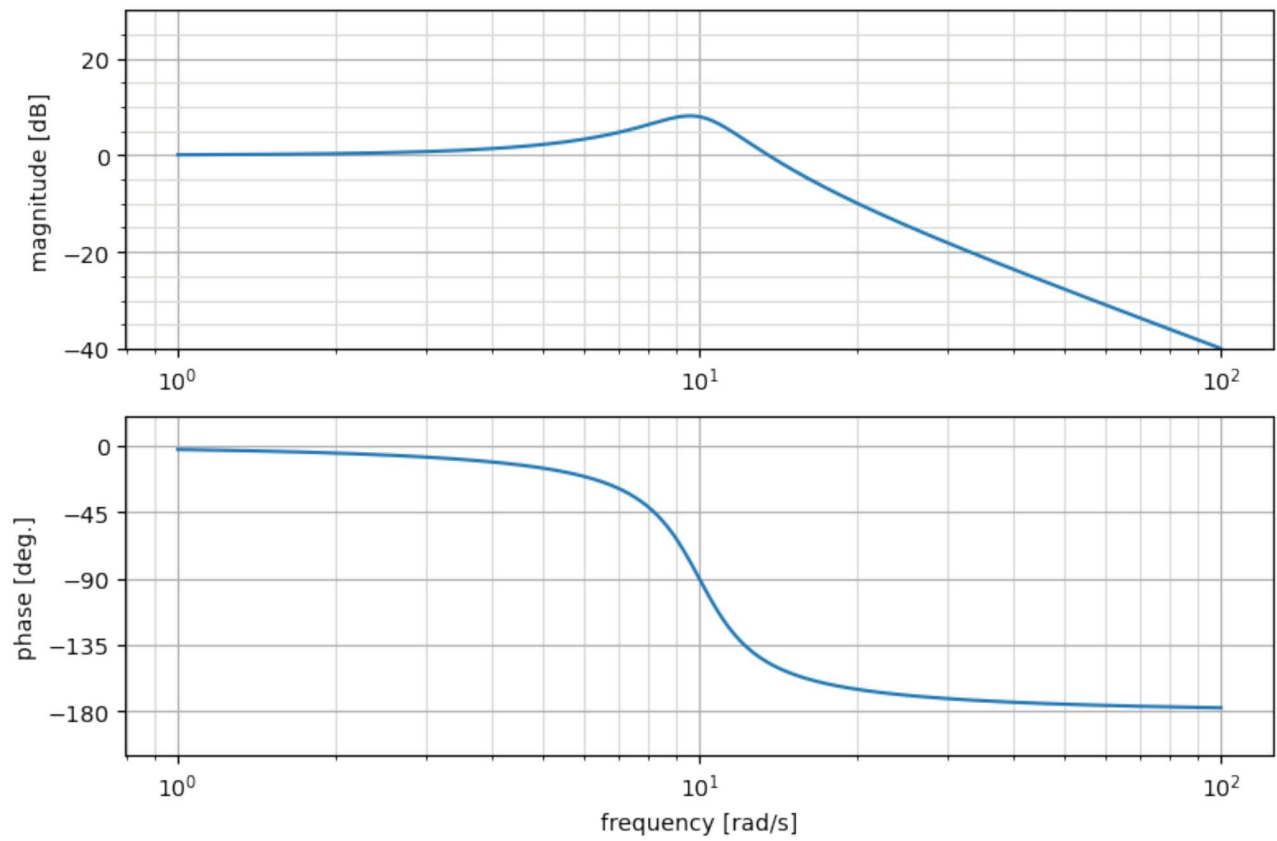


Figure 2: Bode plot

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Question 3 [17 marks]

Consider a system with the transfer function $P(s) = \frac{60}{(s+10)(s+1)(s+3)}$.

- a) Explain why $P_a(s) = \frac{6}{(s+1)(s+3)}$ is a suitable reduced-order model of the system. (You may use this model for the subsequent parts of this question.) (2)

Solution: The pole at -10 is more than 5 times further away from the origin than the other poles, so its contribution to the response will die away much faster. A gain $\frac{1}{10}$ will therefore have a similar effect.

- b) The system is required to track a step reference with 95% accuracy. Your colleague suggests implementing this by putting the system in closed loop with a controller of the form $G_1(s) = \frac{k}{s}$. Explain one advantage, and one disadvantage of this option. You may want to support your answer with a rough sketch of the root locus. (3)

Solution:

Advantage: the integrator will increase the type number of the system, decreasing the position tracking error to zero.

Disadvantage: with another pole at the origin, the system could become unstable at higher gains, while the original model has infinite gain margin.

- c) Another colleague suggests a lead compensator, $G_2(s) = \frac{6.51(s+4.5)}{(s+10.5)}$.

- i) Show that this compensator places the dominant poles of the closed-loop system at $-4 \pm j4$. What will the settling time to 5% of the final value, and damping ratio of the system be if this compensator is used? (5)

Solution: substituting $-4 + j4$ into the combined compensator x plant transfer function gives -1, showing that this is a closed-loop pole.

This pole pair gives a damping ratio of 0.79, a natural frequency of 5.1 and hence a settling time of 0.87 seconds.

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ii) Show that this compensator does not satisfy the tracking specification (2)
Solution: the gain of the forward path must be at least 19 to track a step input to 95%. The gain of this system is only 5.57.

d) Suggest an alternative lead compensator design that places the dominant poles at the same locations as in part (c), but which can provide the required accuracy. (You do not need to reach a working design in this question, but you must show the steps to arrive at another design, and some assessment of whether it works.) (5)

Solution: placing the zero further away will lead to a compensator with a larger gain contribution. Students must state this, and then (1) pick a suitable pole location, and (2) find the required gain. They must then check whether the gain of the overall design is as required.

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QUESTION 4 [18 marks]

Investigate the closed loop stability (with unity gain, negative feedback) of $L(s) = \frac{8(s+6)}{s^2-4s}$.

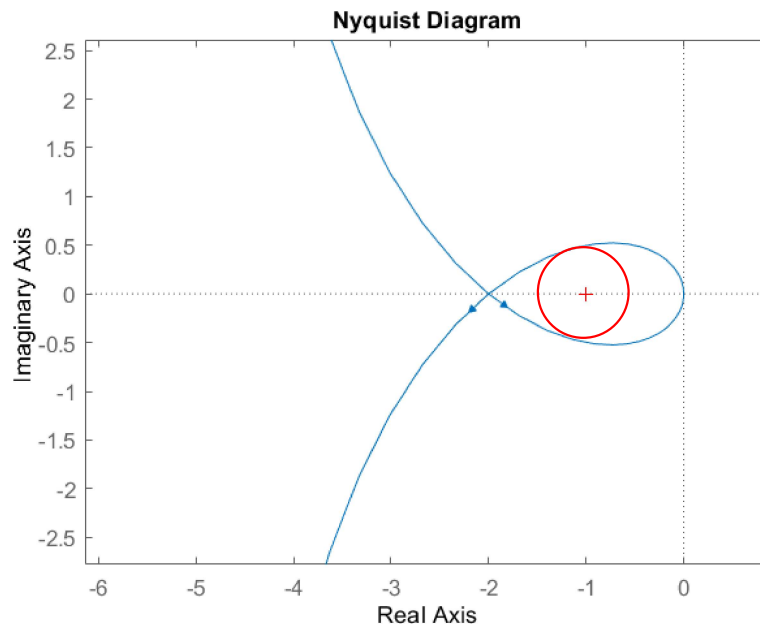
a) How many right-hand plane open loop poles does the system have? (1)

One (plus one on the $j\omega$ axis).

b) Draw the Bode plot on attached semilog paper, using asymptotic values supported by calculations at $\omega = (5, 7, 9)$ rad/s. Make sure that you get the phase in the correct quadrant. (5)

ω	$\left \frac{48 + 8j\omega}{-\omega^2 - 4j\omega} \right $	$\arg \frac{48 + 8j\omega}{-\omega^2 - 4j\omega}$	$L(j\omega)$
0		-270	
5	5.8 dB	-179	-1.95 - 0.04i
7	2.3 dB	-160	-1.23 - 0.44i
9	-0.2 dB	-148	-0.82 - 0.52i
∞		-90	

c) Hence draw the full Nyquist plot with appropriate scaling. Explain clearly how you deal with singularities on the $j\omega$ axis. (4)



Closure is at negative infinity ($L(s) = \frac{8(s+6)}{s^2-4s} = \lim_{\epsilon \rightarrow 0} \frac{8(s+6)}{(s+\epsilon)(s-4)}$). Show closure clearly

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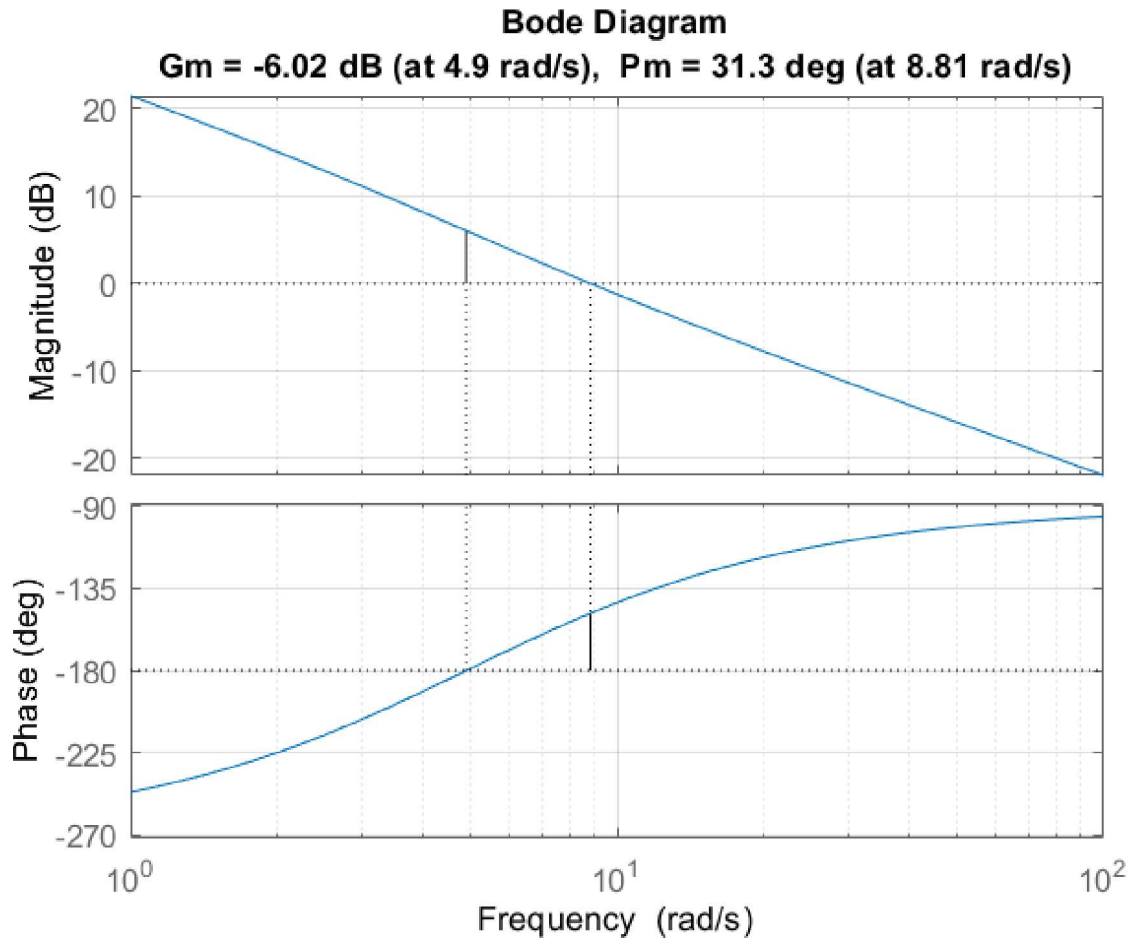
-90	-135	-180	-225	-270
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1	2	3	4	5	7	9	10
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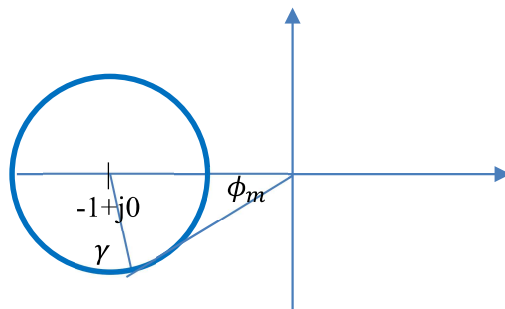
- d) Is the closed loop system stable or not? If it is stable, what are the gain, phase and robust stability margins? If it is unstable, what could be done to stabilise it? (4)

Closed loop system is stable as $P = 1$, $N = -1$, $\Rightarrow Z = 0$. From the Bode plot, $gm = -5.8$ dB (negative because of conditional stability), $pm = 32^\circ$. From Nyquist plot, $\gamma = 0.5$.



- e) Make a sketch and show that a robust stability margin of γ (arithmetic) implies a phase margin of at least $\phi_m \geq 2 \arcsin \frac{1}{2\gamma}$. Hence find the minimum phase margin corresponding to a specification of $\left| \frac{1}{1+L} \right| \leq 6$ dB. (4)

From isosceles triangle with sides, 1, 1, and γ , form two right angled triangles. Then $\frac{\phi_m}{2} = \sin \frac{\gamma}{2} \dots$



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QUESTION 5 [16 marks]

Consider the continuous time plant, $P(s) = \frac{2e^{-0.1s}}{s/3+1}$. Design a continuous time controller, $G(s)$, to achieve the following specifications. Show your design steps clearly on the Inverse Nichols chart. The critical frequency for the robust stability calculations is at $\omega = 4$ rad/s.

i) Zero steady state error to a unit step output disturbance

ii) $\left| \frac{1}{1+L} \right| \leq -15$ dB $\omega \leq 0.6$ rad/s

iii) $\left| \frac{1}{1+L} \right| \leq 3$ dB $\forall \omega$

1) $G_1 = \frac{k}{s}$, $\arg \frac{P(j0.6)}{j0.6} = -104.7^\circ$, $\left| \frac{P(j0.6)}{j0.6} \right| = 10.3$ dB. To get to the -15 dB closed loop line requires gain of 5 dB, $k = 1.8$. (4)

2) $\arg \frac{1.8}{j4} P(j4) = -166.0^\circ$, $\left| \frac{1.8}{j4} P(j4) \right| = -5.4$ dB. Need phase lead of 31° at $\omega = 4$. $G = \frac{1.8(s/6.7+1)}{s}$ (6)

3) Show the specifications and final $L(j\omega)$ on the INC (6)

ω	$ P $	$\arg P$	$ G $	$\arg G$	$ L $	$\arg L$
0.4	5.9	-14.7	9.6	-84.9	15.4	-99.6
1	5.6	-24.2	5.2	-81.5	10.8	-105.7
4	1.6	-76.0	-5.6	-59.2	-4.0	-135.2
10	-4.8	-130.6	-9.8	-33.8	-14.6	-164.4

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QUESTION 6 [12 marks]

Given the continuous time plant, $P_s(s) = \frac{1}{s/6+1}$, a discrete time PI controller, $G_z(w) = \frac{k(w/\alpha+1)}{w}$ has been partially designed with $G_{z1}(w) = \frac{9.5}{w}$. $L_1 = P_s(w)G_{z1}(w)$ (without sampling effect) is illustrated on the attached inverse Nichols chart along with the design specifications.

- i) Allocate 10° for sampling effects at the critical frequency $\omega = 12$ rad/s and find corner frequency, α , of the controller zero, $(w/\alpha + 1)$ to meet the robust stability specification. (4)

At $\omega = 12$ rad/s, require $153-135+10=28$ degrees. $\alpha = \frac{12}{\tan 28^\circ} = 22.6$ rad/s

- ii) Specify the sampling time. (1)

$\frac{12T}{2} = \tan 10^\circ$ giving, $T = \frac{2}{12} \tan 10^\circ = 0.029$ seconds.

- iii) By calculating the magnitude and phase effects of the controller zero and (plant) sampling effect at $\nu = [6, 12, 24]$ rad/s, illustrate the final $L_{zz}(j\nu)$ on the inverse Nichols chart. (You should not need to calculate plant gains etc, as L_1 is shown at these frequencies.) (4)

ν	$\left \frac{w}{22.6} + 1 \right $	$\arg\left(\frac{w}{22.6} + 1\right)$	$ 1-wT/2 $	$\arg\left(1 - \frac{wT}{2}\right)$	$ total $	$\arg(total)$
6	0.3	14.9	0.0	-5.0	0.3	9.9
12	2.2	28.0	0.1	-10.0	2.3	18.0
24	3.3	46.7	0.5	-19.4	3.8	27.3

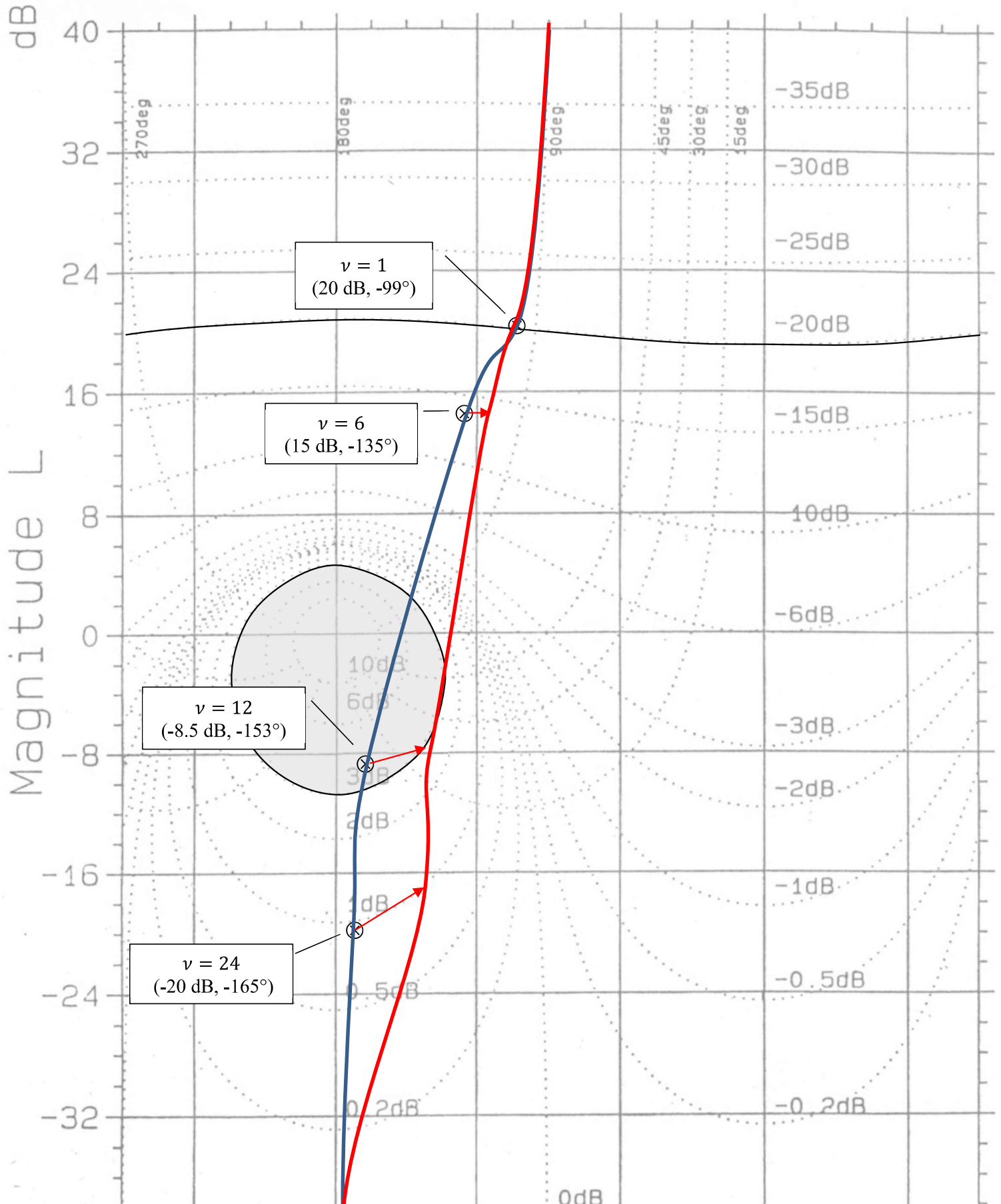
- iv) Find the controller difference equation, $u_{i+1} = \dots$. (3)

$$G_z(z) = G_z(w) \Big|_{w=\frac{2(z-1)}{T(z+1)}} = \frac{0.5599z - 0.28082}{z - 1}$$

$$u_{i+1} = u_i + 0.5599 e_{i+1} - 0.2808 e_i$$

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Inverse Nichols Chart



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Question 7 [8 marks]

Design an observer for the system, $P(s) = \frac{s+5}{s^3+2s^2+3s+4}$, with estimator poles at $s_{1,2} = -1 \pm j$ and $s_3 = -1$. (8)

Observer characteristic equation is: $\Lambda(s) = (s^2 + 2s + 2)(s + 1) = s^3 + 3s^2 + 3s + 2$

$$\text{State Space form} \begin{bmatrix} \begin{bmatrix} -2 & 1 & 0 \\ -3 & 0 & 1 \\ -4 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix} \end{bmatrix}$$

$$\text{Require} \begin{bmatrix} -2 - \ell_2 \\ -3 - \ell_1 \\ -4 - \ell_0 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \\ -2 \end{bmatrix},$$

$$L = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

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Laplace and Z-transforms.

$Z\{y(iT)\}$	$y(t)$	$L\{y(t)\}$ or $P_s(s)$	$P_z(z) = \frac{z-1}{z} \mathcal{Z} \left\{ \left(\mathcal{L}^{-1} \left\{ \frac{P_s(s)}{s} \right\} \right)_{y=iT} \right\}$
$\frac{z}{z-1}$	$\sigma(t)$	$\frac{1}{s}$	$\frac{T}{z-1}$
$\frac{Tz}{(z-1)^2}$	t	$\frac{1}{s^2}$	$\frac{T^2 / 2(z+1)}{(z-1)^2}$
$\frac{z}{z-e^{-aT}}$	e^{-at}	$\frac{1}{s+a}$	$\frac{(1-e^{-aT})/a}{z-e^{-aT}}$
$\frac{Tze^{-aT}}{(z-e^{-aT})^2}$	te^{-at}	$\frac{1}{(s+a)^2}$	$\frac{(1-e^{-aT}(1+aT))z + e^{-aT}(e^{-at}-1+aT)}{(z-e^{-aT})^2 a^2}$
$\frac{ze^{-at} \sin bT}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$	$e^{-at} \sin bt$	$\frac{b}{(s+a)^2 + b^2}$	$\frac{(n_1 z + n_0)b / (a^2 + b^2)}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$ $n_1 = 1 - e^{-aT} \left(\cos bT + \frac{a}{b} \sin bT \right)$ $n_0 = e^{-aT} \left(e^{-aT} - \cos bT + \frac{a}{b} \sin bT \right)$
$\frac{z^2 - ze^{-at} \cos bT}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$	$e^{-at} \cos bt$	$\frac{s+a}{(s+a)^2 + b^2}$	$\frac{(n_1 z + n_0)a / (a^2 + b^2)}{z^2 - 2ze^{-aT} \cos bT + e^{-2aT}}$ $n_1 = 1 - e^{-aT} \left(\cos bT - \frac{b}{a} \sin bT \right)$ $n_0 = e^{-aT} \left(e^{-aT} - \cos bT - \frac{b}{a} \sin bT \right)$

Property	Time domain	z Transform
Linearity	$au(t) + bv(t)$	$au(z) + bv(z)$
Real Translation	$u(t - kT)$	$z^{-k}u(z)$
Initial Value Theorem	$\lim_{t \rightarrow 0} u(t)$	$\lim_{z \rightarrow \infty} u(z)$
Final Value Theorem	$\lim_{t \rightarrow \infty} u(t)$	$\lim_{z \rightarrow 1} (1 - z^{-1})u(z)$